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Batch : CS4

Division : C42

EDS Practical :04

I. Pandas - series creation and manipulation:

Batch: C

Write a Python program that takes a list of numbers from the user, creates a Pandas series from it, and then calculates the mean of even and odd numbers separately using the groupby and mean()

operations.

Hint: You can group the Series based on whether each number is even or odd.

Input Format:

- The user should enter a list of numbers separated by space when prompted.

Output Format:

- The program should display the mean of even and

odd numbers separately.

•

Code:

Each mean value should be displayed with a label indicating whether it corresponds to even or odd numbers.

```
import pandas as pd
```

- Take inputs from the user to create a list of numbers `numbers = list(map(int, input().split()))`
- Create a Pandas series from the list of numbers `series = pd.Series(numbers)`
- Grouping by even and odd numbers and calculating the mean

```
grouped = series.groupby(series % 2 == 0).mean()
```

- Display the mean of even and odd numbers with labels

```
grouped.index = ['Even' if is_even else 'Odd' for is_even in grouped.index]
```

```
print("Mean of even and odd numbers:") print(grouped)
```

The screenshot displays the execution results of a program. At the top, performance metrics are shown: Average time is 0.007 s (7.00 ms) and Maximum time is 0.017 s (17.00 ms). Test results indicate that 3 out of 3 shown test cases and 3 out of 3 hidden test cases passed. Below this, 'Test case 1' is detailed, showing an execution time of 17 ms and a 'Debug' button. The test case compares the 'Expected output' with the 'Actual output'. Both outputs are identical, showing the sequence of numbers 1 through 10, followed by the calculated means for even and odd numbers (5.0 for odd, 6.0 for even), and the data type 'float64'.

Expected output	Actual output
1 2 3 4 5 6 7 8 9 10	1 2 3 4 5 6 7 8 9 10
Mean of even and odd numbers:	Mean of even and odd numbers:
Odd 5.0	Odd 5.0
Even 6.0	Even 6.0
dtype: float64	dtype: float64

At the bottom, there are tabs for 'Terminal' and 'Test cases'.

Output:

1.

I. Dictionary to dataframe:

A dictionary of lists has been provided to you in the editor. Create a DataFrame from the dictionary of lists and perform the listed operations, then display the DataFrame before and after each manipulation.

Create the DataFrame:

- Convert the dictionary to a Pandas DataFrame.

Add a new row:

- Take inputs from the user for the new row data (name, age).
- Add the new row to the DataFrame.

- Display the DataFrame after adding the new row.

Modify a row:

- Modify a specific row by changing the age. Take the row index and new age value from the user.
- Display the DataFrame after modifying the row.

Delete a row:

- Take the row index to be deleted from the user.
- Remove the specified row.
- Display the DataFrame after deleting the row.

Add a new column:

- Add a column Gender with values taken from the user.
- Display the DataFrame after adding the new column.

Modify a column:

- Convert names to uppercase.
- Display the DataFrame after modifying the column.

Delete a column:

- Remove the Age column.
- Display the DataFrame after deleting the column. Code: `import pandas as pd`

- Provided dictionary of lists `data = {`

```
'Name': ['Alice', 'Bob', 'Charlie'], 'Age': [25, 30, 35],
```

```
}
```

- Convert the dictionary to a DataFrame `df = pd.DataFrame(data)`

- Display the original DataFrame `print("Original DataFrame:") print(df)`

- Adding a new row

```
new_name = input("New name: ")
```

```
new_age = int(input("New age: ")) df.loc[len(df)] = [new_name, new_age]
```

- Display the DataFrame after adding a new row `print("After adding a row:\n",df)`

- Modifying a row

```
row_to_modify = int(input("Index of row to modify: ")) new_age_value = int(input("New age: "))  
df.at[row_to_modify, "Age"]
```

```
= new_age_value
```

- Display the DataFrame after modifying a row `print("After modifying a row:")`

```
print(df)
```

- Deleting a row

```
row_to_delete = int(input("Index of row to delete: "))
```

```
df = df.drop(index=row_to_delete).reset_index(drop=True)
```

- Display the DataFrame after deleting a row `print("After deleting a row:")`

```
print(df)
```

- Adding a new column

```
genders=input("Enter genders separated by space: ").split() df["Gender"] = genders
```

- Display the DataFrame after adding a new column `print("After adding a new column:")`

```
print(df)
```

- Modifying a column `df["Name"]=df["Name"].str.upper()`
- Display the DataFrame after modifying a column `print("After modifying a column:")`

`print(df)`

- Deleting a column `df=df.drop(columns=["Age"])`
- Display the DataFrame after deleting a column `print("After deleting a column:")`

`print(df)` Output:

I. Student Information:

Write a program to read a text file containing student information (name, age, and grade) using Pandas.

Perform the following tasks:

- Display the first five rows of the data frame.
- Calculate the average age of the students (limit the average age up to 2 decimal places).
- Filter out the students who have a grade above a certain threshold (consider the threshold grade is 'B').

Code:

```
import pandas as pd
```

- Read the text file into a DataFrame `file = input()`

```
data = pd.read_csv(file, sep="\s+", header=None, names=["Name", "Age", "Grade"])
```

- write your code here.. `df=pd.DataFrame(data)` `print("First five rows:")` `print(df.head(5))`

```
print("Average age:",df["Age"].mean().round(2)) print("Students with a grade up to B")
```

```
filtered_df=df[df['Grade'].isin(['A','B'])] print(filtered_df)
```

Output:

I. Month with the Highest Total Sales:

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Group the data by Month and calculate the total sales for each month.
- Find the month with the highest total sales and display it.
- Also, display the total sales for the best month. Note:

The data cannot be displayed in the file. You can refer to the sample data provided for insights.

Code:

- Prompt the user for the file name `file_name = input()`

- Load the data

```
df = pd.read_csv(file_name)
```

```
df["Date"] = pd.to_datetime(df["Date"]) df["Month"] = df["Date"].dt.to_period("M") df["Total Sales"] =  
df["Quantity"] * df["Price"]
```

```
monthly_sales = df.groupby("Month")["Total Sales"].sum()
```

- Find the month with the highest total sales `best_month = monthly_sales.idxmax()` `highest_sales = monthly_sales.max()`

```
print(f"Best month: {best_month}") print(f"Total sales:
```

```
${highest_sales:.2f}")
```

The screenshot displays a test runner interface with the following components:

- Performance Metrics:**
 - Average time: 0.018 s (17.67 ms)
 - Maximum time: 0.035 s (35.00 ms)
- Test Results Summary:**
 - 1 out of 1 shown test case(s) passed
 - 2 out of 2 hidden test case(s) passed
- Test Case Details (Test case 1):**
 - Duration: 35 ms
 - Buttons: Debug, [List Icon], [File Icon]
 - Expected output:**
 - sales_data.csv
 - Best month: 2025-01
 - Total sales: \$1210.00
 - Actual output:**
 - sales_data.csv
 - Best month: 2025-01
 - Total sales: \$1210.00
- Bottom Bar:**
 - Terminal (active)
 - Test cases

I. Best Selling Product:

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Find the product that sold the most in terms of quantity sold.
- Display the product that sold the most and the total quantity sold for that product.

Code:

```
import pandas as pd
```

```
#
```

```
file_name = input()
```

- Load the data

```
df = pd.read_csv(file_name)
```

```
product_sales = df.groupby("Product")["Quantity"].sum()
```

- Find the product with the highest total quantity sold best_product

```
= product_sales.idxmax() highest_quantity = product_sales.max()
```

- Display the result

```
print(f"Best selling product: {best_product}") print(f"Total quantity sold: {highest_quantity}")
```

 Output:

I. City that Sold the Most Products:

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Group the data by City and calculate the total quantity of products sold for each city.
- Find the city that sold the most products (based on the total quantity sold).

Code:

```
import pandas as pd
```

- Prompt the user for the file name

```
file_name = input()
```

- Load the data

```
df = pd.read_csv(file_name)
```

```
city_sales = df.groupby("City")["Quantity"].sum()
```

```
best_city = city_sales.idxmax()
```

- Display the result

```
print(f"City sold the most products: {best_city}")
```

 Output:

The screenshot displays a test runner interface with the following components:

- Performance Metrics:**
 - Average time:** 0.019 s (18.67 ms)
 - Maximum time:** 0.038 s (38.00 ms)
- Test Results Summary:**
 - 1 out of 1 shown test case(s) passed
 - 2 out of 2 hidden test case(s) passed
- Test Case Details (Test case 1):**
 - Duration:** 38 ms
 - Buttons:** Debug, [List Icon], [File Icon]
 - Expected output:** sales_data.csv
 - Actual output:** sales_data.csv
 - City sold the most products:** Los Angeles

I. Most Frequently Sold Product Pairs:

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

Code:

The CSV file contains the following columns: Date, Product, Quantity, Price, and City.

◆

◆

For each date, find all pairs of products that were sold together (i.e., two products sold on the same date).

◆

Output the product pair/s that was sold most frequently.

```
import pandas as pd
```

```
from itertools import combinations from collections import Counter
```

- Prompt user to input the file name `file_name = input()`
- Read data from the specified CSV file `df = pd.read_csv(file_name)`
- write the code `date_products = {}`

```
for date, group in df.groupby('Date'):
```

```
    products = group['Product'].unique() if len(products) > 1: date_products[date] = products
```

```
grouped = df.groupby("Date")["Product"].apply(list)
```

```
#
```

```
GENERATE ALL PRODUCT PAIRS FOR EACH CONSTRUCTIONS
```

```
pair_counter = Counter()
```

```
for products in date_products.values(): pairs = combinations(sorted(products), 2)
```

```
#
```

```
sort to ensure consistency pair_counter.update(pairs)
```

```
#
```

```
find the most common products pairs if pair_counter: max_count = max(pair_counter.values())
```

#

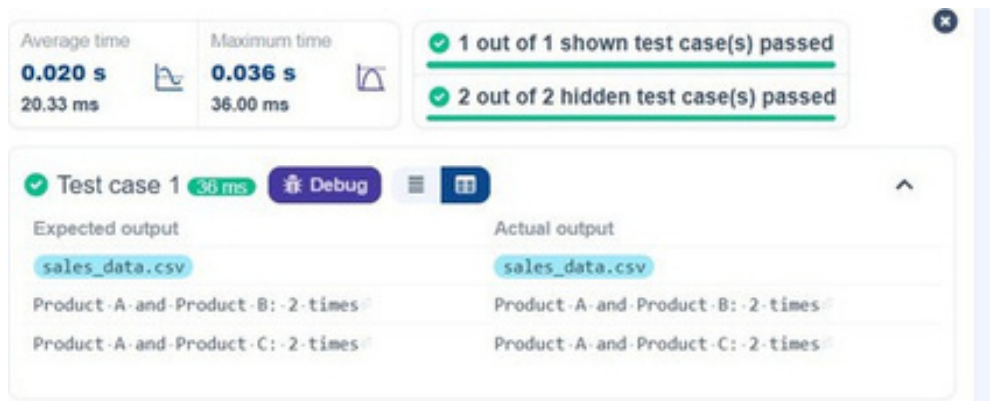
Output the most frequent product pairs for pair, count in pair_counter.items():

if count == max_count:

print(f"{pair[0]} and {pair[1]}: {count} times")

else:

print("No product pairs found.") Output:



I. Titanic Dataset Analysis and Data Cleaning:

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset. For each question, perform necessary data cleaning, transformations, and calculations as required.

1. Display the first 5 rows of the dataset.
2. Display the last 5 rows of the dataset.
3. Get the shape of the dataset (number of rows and columns).
4. Get a summary of the dataset (using `.info()`).
5. Get basic statistics (mean, standard deviation,

etc.) of the dataset using `.describe()`.

1. Check for missing values and display the count of missing values for each column.
2. Fill missing values in the 'Age' column with the median age.
3. Fill missing values in the 'Embarked' column with the most frequent value (`mode()`).
4. Drop the 'Cabin' column due to many missing values.

1. Create a new column, 'FamilySize' by adding the 'SibSp' and 'Parch' columns.

PassengerId

Survive

Pclass :

Name

Sex :

Aa €

SibSi

Parc

Ticke

The Titanic dataset contains columns as shown below,

	F

Code:

```
import pandas as pd
import numpy as np
```

#

Load the Titanic dataset

```
data = pd.read_csv('Titanic-Dataset.csv')
```

1.

#

l. Display the first 5 rows of the dataset `print(data.head())`

1.

#

l. Display the last 5 rows of the dataset `print(data.tail())`

1.

#

l. Get the shape of the dataset `print(data.shape)`

1.

#

I. Get a summary of the dataset (info) `print(data.info())`

1.

#

I. Get basic statistics of the dataset `print(data.describe())`

1.

#

I. Check for missing values `print(data.isnull().sum())`

1.

#

I. Fill missing values in the 'Age' column with the median age

```
data['Age'].fillna(data['Age'].median(), inplace=True)
```

1.

#

I. Fill missing values in the 'Embarked' column with the mode

```
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
```

1.

#

I. Drop the 'Cabin' column due to many missing values `data.drop(columns=['Cabin'], inplace=True)`

1.

#

I. Create a new column 'FamilySize' by adding 'SibSp' and 'Parch' `data['FamilySize'] = data['SibSp'] + data['Parch']` Output:

I. Titanic Dataset Analysis and Data Cleaning – 2:

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Create a new column 'IsAlone' which is 1 if the passenger is alone (FamilySize = 0), otherwise 0.
2. Convert the 'Sex' column to numeric values (male: 0, female: 1).
3. One-hot encode the 'Embarked' column, dropping the first category.
4. Get the mean age of passengers.

1. Get the median fare of passengers.
2. Get the number of passengers by class.
3. Get the number of passengers by gender.
4. Get the number of passengers by survival status.
5. Calculate the survival rate of passengers.
6. Calculate the survival rate by gender.

PassengerId

Survive

Pclass :

Name

Se :

Age

SibSp

Parc

Ticket

The Titanic dataset contains columns as shown below,

	F

Code:

```
import pandas as pd
import numpy as np
```

#

Load the Titanic dataset

```
data = pd.read_csv('Titanic-Dataset.csv') data['FamilySize'] = data['SibSp'] + data['Parch']
```

1.

#

I. Create a new column 'IsAlone' (1 if alone, 0 otherwise)

```
data['IsAlone'] = np.where(data['FamilySize'] == 0, 1, 0)
```

1.

#

I. Convert 'Sex' to numeric (male: 0, female: 1)

```
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
```

1.

#

I. One-hot encode the 'Embarked' column

```
data = pd.get_dummies(data, columns=['Embarked'])
```

1.

#

I. Get the mean age of passengers mean_age = data['Age'].mean() print(mean_age)

1.

#

I. Get the median fare of passengers median_fare = data['Fare'].median() print(median_fare)

1.

#

I. Get the number of passengers by class print(data['Pclass'].value_counts()
)

1.

#

1. Get the number of passengers by gender `print( data['Sex'].value_counts())`

1.

#

1. Get the number of passengers by survival status `print( data['Survived'].value_counts())`

2.

#

1. Calculate the survival rate `survival_rate = data['Survived'].mean() print(format(survival_rate))`

1.

#

1. Calculate the survival rate by gender

`survival_by_gender = data.groupby('Sex') ['Survived'].mean() print( survival_by_gender)` Output:

Average time
0.089 s
89.00 ms

Maximum time
0.089 s
89.00 ms

1 out of 1 shown test case(s) passed

Test case 1 89 ms Debug

Expected output

Actual output

29.69911764705882

29.69911764705882

14.4542

14.4542

3...491

3...491

1...216

1...216

2...184

2...184

Name: Pclass, dtype: int64

Name: Pclass, dtype: int64

0...577

0...577

1...314

1...314

Name: Sex, dtype: int64

Name: Sex, dtype: int64

0...549

0...549

1...342

1...342

Name: Survived, dtype: int64

Name: Survived, dtype: int64

0.3838383838383838

0.3838383838383838

Sex

Sex

0...0.188908

0...0.188908

1...0.742038

1...0.742038

I. Titanic Dataset Analysis and Data Cleaning – 3:

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to

answer the following questions based on the dataset.

1. Calculate the survival rate by class.
2. Calculate the survival rate by embarkation location (Embarked_S).
3. Calculate the survival rate by family size

(FamilySize).

1. Calculate the survival rate by being alone (IsAlone).
2. Get the average fare by passenger class (Pclass).
3. Get the average age by passenger class (Pclass).
4. Get the average age by survival status (Survived).
5. Get the average fare by survival status (Survived).
6. Get the number of survivors by class (Pclass).
7. Get the number of non-survivors by class (Pclass).

PassengerId

Survived

Pclass

Name

Sex

Age

Siblings

Parcels

Ticke

The Titanic dataset contains columns as shown below,

	F

Code:

```
import pandas as pd import numpy as np
```

#

Load the Titanic dataset

```
data = pd.read_csv('Titanic-Dataset.csv') data['FamilySize'] = data['SibSp'] + data['Parch']
```

```
data['IsAlone'] = np.where(data['FamilySize'] > 0, 0, 1)
```

```
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
```

#

I. Calculate the survival rate by class `print(data.groupby('Pclass')['Survived'].mean())`

1.

#

I. Calculate the survival rate by embarked location `print(data.groupby('Embarked_S')['Survived'].mean())`

1.

#

I. Calculate the survival rate by family size `print(data.groupby('FamilySize')['Survived'].mean())`

1.

#

I. Calculate the survival rate by being alone `print(data.groupby('IsAlone')['Survived'].mean())`

1.

#

l. Get the average fare by class `print(data.groupby('Pclass')['Fare'].mean())`

1.

#

l. Get the average age by class `print(data.groupby('Pclass')['Age'].mean())`

1.

#

l. Get the average age by survival status `print(data.groupby('Survived')['Age'].mean())`

1.

#

1. Get the average fare by survival status `print(data.groupby('Survived')['Fare'].mean())`

1.

#

1. Get the number of survivors by class `print(data[data['Survived']
== 1]['Pclass'].value_counts())`

1.

#

1. Get the number of non-survivors by class `print(data[data['Survived'] == 0]['Pclass'].value_counts())`
Output:

1. Titanic Dataset Analysis and Data Cleaning – 4:

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Get the number of survivors by gender (Sex).
2. Get the number of non-survivors by gender (Sex).
3. Get the number of survivors by embarkation location (Embarked_S).
4. Get the number of non-survivors by embarkation location (Embarked_S).

1. Calculate the percentage of children ($\text{Age} < 18$) who survived.
2. Calculate the percentage of adults ($\text{Age} \geq 18$) who survived.
3. Get the median age of survivors.
4. Get the median age of non-survivors.
5. Get the median fare of survivors.
6. Get the median fare of non-survivors.

PassengerId

Survived

Pclass :

Name

Sex :

Age

Siblings

Par

Ticke

The Titanic dataset contains columns as shown below,

	F

Code:

```
import pandas as pd import numpy as np
```

```
#
```

Load the Titanic dataset

```
data = pd.read_csv('Titanic-Dataset.csv')
```

```
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
```

1.

```
#
```

1. Get the number of survivors by gender

```
survivors_by_gender = data[data['Survived'] == 1] ['Sex'].value_counts()  
print(survivors_by_gender)
```

1.

#

1. Get the number of non-survivors by gender

```
non_survivors_by_gender = data[data['Survived'] == 0] ['Sex'].value_counts()  
print(non_survivors_by_gender)
```

1.

#

1. Get the number of survivors by embarked location

```
survivors_by_embarked_s = data[data['Survived'] == 1] ['Embarked_S'].value_counts()
```

```
print(survivors_by_embarked_s)
```

1.

#

1. Get the number of non-survivors by embarked location non_survivors_by_embarked_s = data[data['Survived'] == 0]['Embarked_S'].value_counts() print(non_survivors_by_embarked_s)

1.

#

1. Calculate the percentage of children (Age < 18) who survived

```
children = data[data['Age'] < 18]
children_survival_rate = children['Survived'].mean()
print(children_survival_rate)
```

1.

#

1. Calculate the percentage of adults (Age >= 18) who survived

```
adults = data[data['Age'] >= 18]
adults_survival_rate = adults['Survived'].mean()
print(adults_survival_rate)
```

1.

#

1. Get the median age of survivors

```
median_age_survivors = data[data['Survived'] == 1]['Age'].median()
print(median_age_survivors)
```

1.

#

I. Get the median age of non-survivors

```
median_age_non_survivors = data[data['Survived'] == 0] ['Age'].median()  
print(median_age_non_survivors)
```

1.

#

I. Get the median fare of survivors

```
median_fare_survivors = data[data['Survived'] == 1] ['Fare'].median() print(median_fare_survivors)
```

1.

#

I. Get the median fare of non-survivors

```
median_fare_non_survivors = data[data['Survived'] == 0] ['Fare'].median()
```

```
print(median_fare_non_survivors) Output:
```

Average time

0.116 s

116.00 ms



Maximum time

0.116 s

116.00 ms



1 out of 1 shown test case(s) passed

Test case 1

116 ms

Debug



Expected output

female...233

male...109

Name: Sex, dtype: int64

male...468

female...81

Name: Sex, dtype: int64

1...217

0...125

Name: Embarked_S, dtype: int64

1...427

0...122

Name: Embarked_S, dtype: int64

0.5398230088495575

0.3810316139767055

28.0

28.0

Actual output

female...233

male...109

Name: Sex, dtype: int64

male...468

female...81

Name: Sex, dtype: int64

1...217

0...125

Name: Embarked_S, dtype: int64

1...427

0...122

Name: Embarked_S, dtype: int64

0.5398230088495575

0.3810316139767055

28.0

28.0